2005 Nissan 350Z VQ35DE — Engine harness print pack

Motor-control and body loom · ECM MEC61-510 · cards with FSM values

Card · color = outline · 5V amber / 12V red / GND green · ECM Inverted L/R · cards FSM T.S. · FSM EC 2005 · OEM · not a service manual

EC-123 INVERTED — MEC61-510

pin # top · wire code bottom · border = wire · red = bench · Inverted L/R

TOP | 121 on the LEFT

LEVER — B=black · W=white · R=red · G=green · L=blue · Y=yellow · OR=orange · P=pink · PU=purple · GY=gray · BR=brown · SB=sky · LG=lt.green — LEVER

121	120	119	113	112	111	110	109	108	107	106	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	5	4
R/W	P	R/W	LG/B	•	GY/L	•	W/B	B/R	GY/R	G/OR	P/L	•	P	SB	W/G	P	L/W	W/L	R/L	W	LG/B	•	•	•	•	SB	R/Y	R/B	GY	L/Y	L/B
12V	12V	12V																													
OFF: 11–14 V	ON: 11–14 V	ON: 11–14 V			Running: 0–1.5 V OFF (aft…		OFF: 0 V ON: 11–14 V	Brake pedal and/or…	pproximate 0–4.8 V Output…	Fully released: 0.5–1.0 …	After the warm-up protocol…		Idle (warm-up condition…	Idle (warm-up condition…	Idle (warm-up condition…	Idle (warm-up condition…	Steering wheel being…	Idle (warm-up condition…	Idle (warm-up condition…	Idle (warm-up condition…	pproximate 2.5 V						BATTERY VOLTAGE (11–14 V…	BATTERY VOLTAGE (11–14 V…	BATTERY VOLTAGE (11–14 V…	pproximate 5 V		
118	117	105	104	103	102	101	100	99	98	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43		3
•	G/W	•	SB	•	BR/Y	L/OR	•	G/Y	G/B	P/B	•	•	•	•	•	•	BR/W	R	Y/G	L/W	•	•	•	•	LG	W/B	B/R			G
	BATTERY VOLTAGE (11–14 V)		OFF: 11–14 V ON: 0–1.…		Shift lever in P or N (A/T),…	Brake pedal fully released…		ASCD steering switch…	Fully released: 0.15–0.6…	After the warm-up protocol…							Vacuum not applied:…	Idle (warm-up condition…	pproximate 0–4.8 V Output…						BATTERY VOLTAGE (11–14 V…	BATTERY VOLTAGE (11–14 V…	BATTERY VOLTAGE (11–14 V…			12V
																															ON: 11–14 V
116	115	114	97	96	95	94	93	92	91	90	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	2	1
B/R	B/W	•	•	•	•	L	•	•	B/OR	LG	•	G/W	•	W/R	PU	B	G	OR	•	L/B	•	L/B	W/L	L	W	•	PU/W	L/R	Y/R	GY/R	B
GND	GND					C/H			5V	5V				5V	5V	5V														GND
Idle: body ground	Idle: body ground								pproximate 5 V	pproximate 5 V		BATTERY VOLTAGE (11–14 V…		pproximate 5 V	pproximate 5 V	pproximate 5 V	Accelerator pedal fully released…	Ignition switch ON (engine…		Idle speed: 0–1.0 V Engine…		0 - pproximate 1.0 V					Idle (warm-up condition…	Idle (warm-up condition…	Idle (warm-up condition…	pproximate 5 V	Idle: body ground
114–121			89	88	87	86	85	84	83	82	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81		1–5
			•	•	•	P	LG	•	G/R	G/Y	•	•	•	L	B/W	R/W	Y	G/R	•	L/W	BR/Y	OR	P/B	LG	P	B/Y	GY/R	GY	G/B		
						C/L			GND	GND				GND	GND	5V										GND					
									Idle: ~0 V	Idle: ~0 V				Idle: ~0 V	Idle: ~0 V	pproximate 5 V	Accelerator pedal fully released…	1.0–4.0 V		Idle speed: 0–1.0 V Engine…	pproximate 0–4.8 V Output…	0 - pproximate 1.0 V				Idle: ~0 V	Idle (warm-up condition…	Idle (warm-up condition…	Idle (warm-up condition…		
			82–113											6–81																	

ECM RAILS · lines leaving the module (engine harness)

pin border = wire color · FSM EC-123

12V	BATT 121=R/W (E108·68G → F102·5H) · IGN 119=R/W · IGN 120=P · VMOT 3 (E8·42)
5V	47 TPS/ETC · 48 EVAP · 49 A/C press · 68 PSP · 90 APP1 · 91 APP2 (separate rails, not one shared 5V)
GND	ECM 1=B · 115=B/W · 116=B/R (→F103·3/2/4→F152) · SNS 66/67/78/82/83 · signals: KNK15 · CKP13 · CAN 86/94 · AFh 2/24

FSM values at ECM terminals

Pin	Value (conditions)	Source
1	Idle: body ground	
2	Approximately 5 V	EC-124
3	ON: 11–14 V	
6	After the warm-up protocol (3,500–4,000 rpm for 1 minute and idle for 1 minute under no load), below 3,600 rpm: 0–1.0 V · Ignition switch ON, engine stopped, or above 3,600 rpm: BATTERY VOLTAGE (11–14 V)	EC-125
8	Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · Engine speed above 1,500 rpm (warm-up condition): 7–12 V	EC-125
9	Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · Engine speed above 1,500 rpm (warm-up condition): 7–12 V	EC-125
10	Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · When revving the engine up to 2,000 rpm quickly (warm-up condition): 7–12 V	EC-126
11	Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · When revving the engine up to 2,000 rpm quickly (warm-up condition): 7–12 V	EC-126
12	Steering wheel being turned: 0.5–4.5 V · Steering wheel not being turned: 0.4–0.8 V	EC-429
13	Idle (warm-up condition): Approximately 1.6 V · Engine speed 2,000 rpm: Approximately 1.4 V · The pulse cycle changes depending on rpm.	EC-126
14	Idle (warm-up condition): 1.0–4.0 V · Engine speed 2,000 rpm: 1.0–4.0 V · The pulse cycle changes depending on rpm.	EC-127
15	Approximately 2.5 V	EC-127
21	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
22	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
23	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
24	Approximately 5 V	EC-127
25	After the warm-up protocol (3,500–4,000 rpm for 1 minute and idle for 1 minute under no load), below 3,600 rpm: 0–1.0 V · Ignition switch ON, engine stopped, or above 3,600 rpm: BATTERY VOLTAGE (11–14 V)	EC-128
32	Vacuum not applied: 1.8–4.8 V · At –26.7 kPa (–200 mmHg, –7.87 inHg): 2.1 to 2.5 V lower than the above value	EC-376
33	Idle (warm-up condition): 1.0–4.0 V · Engine speed 2,000 rpm: 1.0–4.0 V · The pulse cycle changes depending on rpm.	EC-128
34	Approximately 0–4.8 V · Output voltage varies with intake air temperature	EC-128
40	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128
41	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128
42	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128
45	BATTERY VOLTAGE (11–14 V) at idle (accelerator pedal not depressed even slightly, after engine starting) and at about 2,000 rpm (more than 100 seconds after starting the engine)	EC-129
47	Approximately 5 V	EC-129
48	Approximately 5 V	EC-129
49	Approximately 5 V	EC-718
50	Accelerator pedal fully released: more than 0.36 V · Fully depressed: less than 4.75 V	EC-240
51	Ignition switch ON (engine stopped.): Approx. 0.4 V · Idle (engine warmed-up to normal operating temperature.): 0.9–1.2 V · 2,500 rpm (engine warmed-up to normal operating temperature.): 1.5–1.9 V · Idle to about 4,000 rpm: 0.9–1.2 V to Approx. 2.4 V (linear voltage rise)	EC-212
53	Idle speed: 0–1.0 V · Engine speed 2,000 rpm: 0–1.0 V · The pulse cycle changes depending on rpm.	EC-130
55	0 - Approximately 1.0 V	EC-130
60	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-130
61	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-130
62	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-130

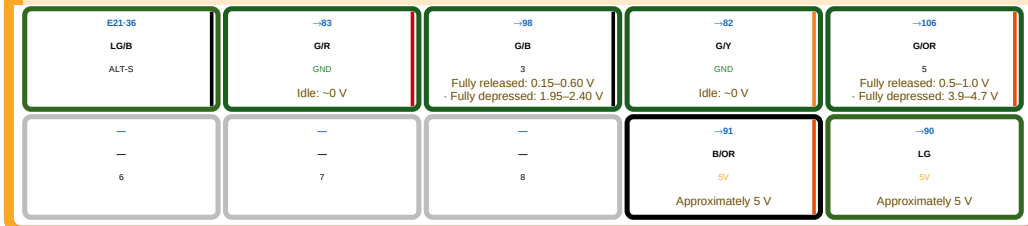
Pin	Value (conditions)	Source
66	Idle: ~0 V	
67	Idle: ~0 V	
68	Approximately 5 V	EC-131
69	Accelerator pedal fully released: less than 4.75 V · Fully depressed: more than 0.36 V	EC-240
70	1.0–4.0 V	EC-718
72	Idle speed: 0–1.0 V · Engine speed 2,000 rpm: 0–1.0 V · The pulse cycle changes depending on rpm.	EC-131
73	Approximately 0–4.8 V · Output voltage varies with engine coolant temperature	EC-131
74	0 - Approximately 1.0 V	EC-131
78	Idle: ~0 V	
79	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-132
80	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-132
81	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-132
82	Idle: ~0 V	
83	Idle: ~0 V	
90	Approximately 5 V	EC-132
91	Approximately 5 V	EC-132
98	Fully released: 0.15–0.60 V · Fully depressed: 1.95–2.40 V	EC-670
99	ASCD steering switch OFF: Approximately 4 V · MAIN switch pressed: Approximately 0 V · CANCEL switch pressed: Approximately 1 V · RESUME/ACCELERATE switch pressed: Approximately 3 V · SET/COAST switch pressed: Approximately 2 V	EC-133
101	Brake pedal fully released: Approximately 0 V · Brake pedal slightly depressed: BATTERY VOLTAGE (11–14 V)	EC-133
102	Shift lever in P or N (A/T), Neutral (M/T): Approximately 0 V · Except above position: BATTERY VOLTAGE (11–14 V)	EC-133
104	OFF: 11–14 V · ON: 0–1.0 V	
106	Fully released: 0.5–1.0 V · Fully depressed: 3.9–4.7 V	EC-670
107	Approximately 0–4.8 V · Output voltage varies with fuel tank temperature	EC-133
108	Brake pedal and/or clutch pedal slightly depressed (M/T), or brake pedal slightly depressed (A/T): Approximately 0 V · Brake and clutch pedals fully released (M/T), or brake pedal fully released (A/T): BATTERY VOLTAGE (11–14 V)	EC-133
109	OFF: 0 V · ON: 11–14 V	
111	Running: 0–1.5 V · OFF (after a few s): 11–14 V	
115	Idle: body ground	
116	Idle: body ground	
117	BATTERY VOLTAGE (11–14 V)	EC-134
119	ON: 11–14 V	
120	ON: 11–14 V	
121	OFF: 11–14 V	

Pedal and switches

E11 ↔ F2

APP path (pedal)

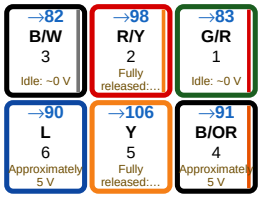
10-pin GY · E113→E11/F2→ECM · EC-523



Pin	Value (conditions)	Source
2	Idle: ~0 V	
3	Fully released: 0.15–0.60 V · Fully depressed: 1.95–2.40 V	EC-670
4	Idle: ~0 V	
5	Fully released: 0.5–1.0 V · Fully depressed: 3.9–4.7 V	EC-670
9	Approximately 5 V	EC-132
10	Approximately 5 V	EC-132

APP pedal E1

6-pin GY · pedal · PAT



Pin	Value (conditions)	Source
1	Idle: ~0 V	
2	Fully released: 0.15–0.60 V · Fully depressed: 1.95–2.40 V	EC-670
3	Idle: ~0 V	
4	Approximately 5 V	EC-132
5	Fully released: 0.5–1.0 V · Fully depressed: 3.9–4.7 V	EC-670
6	Approximately 5 V	EC-132

Stop lamp

Brake switch

E111 (A/T) / E112 (M/T) · ECM 101 L/OR



Pin	Value (conditions)	Source
2	Brake pedal fully released: Continuity should not exist. Brake pedal slightly depressed: Continuity should exist · Brake pedal fully released: Approximately 0 V · Brake pedal slightly depressed: BATTERY VOLTAGE (11–14 V)	EC-639
1	Brake pedal fully released: Continuity should not exist. Brake pedal slightly depressed: Continuity should exist	EC-639

ASCD brake s

2-pin · term2 → ECM 10



Pin	Value (conditions)	Source
1	Brake pedal fully released: Continuity should exist. Brake pedal slightly depressed: Continuity should not exist	EC-639
2	Brake pedal fully released: Continuity should exist. Brake pedal slightly depressed: Continuity should not exist Brake pedal and/or clutch pedal slightly depressed (M/T), or brake pedal slightly depressed (A/T): Approximately 0 V Brake and clutch pedals fully released (M/T), or brake pedal fully released (A/T): BATTERY VOLTAGE (11–14 V)…	EC-639

ASCD clutch

M/T only · in series w



Pin	Value (conditions)	Source
1	Clutch pedal fully released: Continuity should exist. Clutch pedal slightly depressed: Continuity should not exist	EC-639
2	Clutch pedal fully released: Continuity should exist. Clutch pedal slightly depressed: Continuity should not exist Brake pedal and/or clutch pedal slightly depressed (M/T), or brake pedal slightly depressed (A/T): Approximately 0 V Brake and clutch pedals fully released (M/T), or brake pedal fully released (A/T): BATTERY VOLTAGE (11–14 V)…	EC-639

Engine sensors

A/F

Sensor 1 B1

6-pin · F22 · EC-434/529

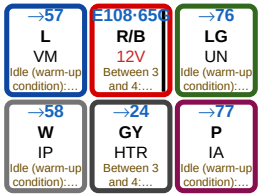


Pin	Value (conditions)	Source
1	Idle (warm-up condition): Approximately 3.1 V	EC-127
2	Idle (warm-up condition): Approximately 2.3 V	EC-127
3	Between 3 and 4: 2.3–4.3 ohm [at 25 °C (77 °F)]. Between 3 or 4 and 1, 2, 5 or 6: Continuity should not exist	EC-439
4	Between 3 and 4: 2.3–4.3 ohm [at 25 °C (77 °F)]. Between 3 or 4 and 1, 2, 5 or 6: Continuity should not exist · Approximately 5 V	EC-439
5	Idle (warm-up condition): Approximately 2.6 V	EC-127
6	Idle (warm-up condition): Approximately 2.3 V	EC-127

A/F

Sensor 1 B2

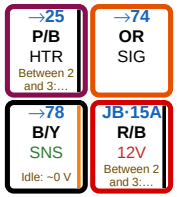
6-pin · F34 · EC-436/531



Pin	Value (conditions)	Source
1	Idle (warm-up condition): Approximately 3.1 V	EC-130
2	Idle (warm-up condition): Approximately 2.3 V	EC-130
3	Between 3 and 4: 2.3–4.3 ohm [at 25 °C (77 °F)]. Between 3 or 4 and 1, 2, 5 or 6: Continuity should not exist	EC-439
4	Between 3 and 4: 2.3–4.3 ohm [at 25 °C (77 °F)]. Between 3 or 4 and 1, 2, 5 or 6: Continuity should not exist · Approximately 5 V	EC-439
5	Idle (warm-up condition): Approximately 2.6 V	EC-130
6	Idle (warm-up condition): Approximately 2.3 V	EC-130

HO2S2 B1

post-cat R · signal EC

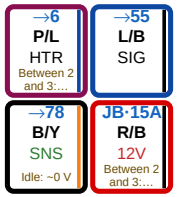


Pin	Value (conditions)	Source
1	0 - Approximately 1.0 V	EC-131
2	Between 2 and 3: 5.0–7.0 ohm [at 25 °C (77 °F)]. Between 1 and 2, 3 or 4, and between 4 and 1, 2 or 3: ∞ ohm (Continuity should not exist)	EC-196
3	Between 2 and 3: 5.0–7.0 ohm [at 25 °C (77 °F)]. Between 1 and 2, 3 or 4, and between 4 and 1, 2 or 3: ∞ ohm (Continuity should not exist) · After the warm-up protocol (3,500–4,000 rpm for 1 minute and idle for 1 minute under no load), below 3,600 rpm: 0–1.0 V · Ignition switch ON, engine stopped, or above 3,600 rpm: BATTERY VOLTAGE (11–14 V)	EC-196
4	Idle: ~0 V	

Engine sensors

HO2S2 B2

post-cat L · signal EC



Pin	Value (conditions)	Source
1	0 - Approximately 1.0 V	EC-130
2	Between 2 and 3: 5.0–7.0 ohm [at 25 °C (77 °F)]. Between 1 and 2, 3 or 4, and between 4 and 1, 2 or 3: ∞ ohm (Continuity should not exist)	EC-196
3	Between 2 and 3: 5.0–7.0 ohm [at 25 °C (77 °F)]. Between 1 and 2, 3 or 4, and between 4 and 1, 2 or 3: ∞ ohm (Continuity should not exist) · After the warm-up protocol (3,500–4,000 rpm for 1 minute and idle for 1 minute under no load), below 3,600 rpm: 0–1.0 V · Ignition switch ON, engine stopped, or above 3,600 rpm: BATTERY VOLTAGE (11–14 V)	EC-196
4	Idle: ~0 V	

MAF

Mass air flow

F25 · 6-pin · EC-207



Pin	Value (conditions)	Source
2	Battery voltage	EC-210
3	Idle: ~0 V	
4	Ignition switch ON (engine stopped.): Approx. 0.4 V · Idle (engine warmed-up to normal operating temperature.): 0.9–1.2 V · 2,500 rpm (engine warmed-up to normal operating temperature.): 1.5–1.9 V · Idle to about 4,000 rpm: 0.9–1.2 V to Approx. 2.4 V (linear voltage rise)	EC-212
5	At 25 °C (77 °F): 1.800–2.200 kohm · Approximately 0–4.8 V · Output voltage varies with intake air temperature	EC-226
6	At 25 °C (77 °F): 1.800–2.200 kohm	EC-226

Knock

sensor

2-pin · via intermediate F14/F229



Pin	Value (conditions)	Source
1	Approximately 2.5 V · Approximately 532–588 kohm [at 20 °C (68 °F)]	EC-127
2	Idle: body ground	

CKP

Crank position

3-pin



Pin	Value (conditions)	Source
1	Battery voltage · Except 0 or ∞ [at 25 °C (77 °F)]	EC-326
2	Idle (warm-up condition): Approximately 1.6 V · Engine speed 2,000 rpm: Approximately 1.4 V · The pulse cycle changes depending on rpm. · Except 0 or ∞ [at 25 °C (77 °F)]	EC-126
3	Except 0 or ∞ [at 25 °C (77 °F)]	EC-328

CMP

Cam phase B1

F4 GY/3 · cam RH · EC-331



Pin	Value (conditions)	Source
1	Except 0 or ∞ [at 25 °C (77 °F)]	EC-338
2	Idle (warm-up condition): 1.0–4.0 V · Engine speed 2,000 rpm: 1.0–4.0 V · The pulse cycle changes depending on rpm. · Except 0 or ∞ [at 25 °C (77 °F)]	EC-128
3	Except 0 or ∞ [at 25 °C (77 °F)]	EC-338

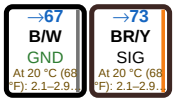
Engine sensors

CMP
Cam phase B2 — F32 B/3 · cam LH · EC-333



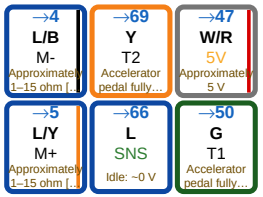
Pin	Value (conditions)	Source
1	Except 0 or ∞ [at 25 °C (77 °F)]	EC-338
2	Idle (warm-up condition): 1.0–4.0 V · Engine speed 2,000 rpm: 1.0–4.0 V · The pulse cycle changes depending on rpm. · Except 0 or ∞ [at 25 °C (77 °F)]	EC-127
3	Except 0 or ∞ [at 25 °C (77 °F)]	EC-338

ECT
Coolant temp — F13 GY/2 · EC-229



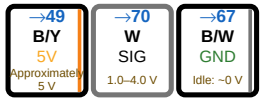
Pin	Value (conditions)	Source
1	At 20 °C (68 °F): 2.1–2.9 kohm · At 50 °C (122 °F): 0.68–1.00 kohm · At 90 °C (194 °F): 0.236–0.260 kohm · Approximately 0–4.8 V · Output voltage varies with engine coolant temperature	EC-232
2	At 20 °C (68 °F): 2.1–2.9 kohm · At 50 °C (122 °F): 0.68–1.00 kohm · At 90 °C (194 °F): 0.236–0.260 kohm	EC-232

ETC
Throttle body DBW — F31 GY/6 · EC-235/464



Pin	Value (conditions)	Source
1	Approximately 5 V	EC-129
2	Accelerator pedal fully released: less than 4.75 V · Fully depressed: more than 0.36 V	EC-240
3	Approximately 1–15 ohm [at 25 °C (77 °F)]	EC-469
4	Accelerator pedal fully released: more than 0.36 V · Fully depressed: less than 4.75 V	EC-240
5	Idle: ~0 V	
6	Approximately 1–15 ohm [at 25 °C (77 °F)]	EC-469

A/C refriger
3-pin · 5V=49 · SIG=70



Pin	Value (conditions)	Source
1	Approximately 5 V	EC-718
2	1.0–4.0 V	EC-718
3	Idle: ~0 V	

EVAP
Pressure sensor — T21 · canister · EC-379



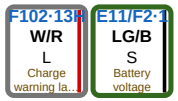
Pin	Value (conditions)	Source
3	Approximately 5 V	EC-129
2	Vacuum not applied: 1.8–4.8 V At –26.7 kPa (–200 mmHg, –7.87 inHg): 2.1 to 2.5 V lower than the above value	EC-376
1	Idle: ~0 V	

PSP
Power steering pressure — 3-pin · 5V=68 · SIG=12 · GND=67



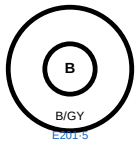
Pin	Value (conditions)	Source
1	Approximately 5 V	EC-131
2	Steering wheel being turned: 0.5–4.5 V · Steering wheel not being turned: 0.4–0.8 V	EC-429
3	Idle: ~0 V	

F20
Alternator S/L — GY/2 · pins 3-4 · SC-22



Pin	Value (conditions)	Source
3	Charge warning lamp should light up	SC-27
4	Battery voltage	SC-27

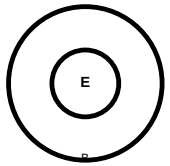
Alternator
B (E202) — B terminal · ring · charge output · SC



Pin	Value (conditions)	Source
1	To ground: Battery voltage. Voltage drop to the battery positive terminal, engine running at idle and warm: Less than 0.2 V	SC-28

Alternator

E (E211) — E terminal · ring · ground · SC-22



F21

Oil pressure — B/3 · DI-34/66



Pin	Value (conditions)	Source
1	Approx. 5 V	DI-45
2	Ignition switch ON (engine stopped): Approx. 1 V · Engine running [when the oil pressure is 80 psi (500 kPa)]: Approx. 3 V	DI-45

F38

EVT pos. B1 — B/3 · Rev-Up (35th M/T *1) · EC-445 ·



Pin	Value (conditions)	Source
1	Except 0 or ∞ [at 25 °C (77 °F)]	EC-453
2	Idle speed: 0–1.0 V · Engine speed 2,000 rpm: 0–1.0 V · The pulse cycle changes depending on rpm. · Except 0 or ∞ [at 25 °C (77 °F)]	EC-130
3	Except 0 or ∞ [at 25 °C (77 °F)]	EC-453

F42

EVT pos. B2 — GY/3 · Rev-Up (35th M/T *1) · EC-447 ·



Pin	Value (conditions)	Source
1	Except 0 or ∞ [at 25 °C (77 °F)]	EC-453
2	Idle speed: 0–1.0 V · Engine speed 2,000 rpm: 0–1.0 V · The pulse cycle changes depending on rpm. · Except 0 or ∞ [at 25 °C (77 °F)]	EC-131
3	Except 0 or ∞ [at 25 °C (77 °F)]	EC-453

F242

Oil temp — GY/2 · Rev-Up 35th M/T (*1) · sub-harn



Pin	Value (conditions)	Source
G	Idle: ~0 V	

F35

PNP (M/T) — B/2 · M/T · ECM 102 BR/Y · EC-646 · PG



Pin	Value (conditions)	Source
1	Neutral: Continuity Yes. Except neutral: Continuity No · Shift lever in P or N (A/T), Neutral (M/T): Approximately 0 V · Except above position: BATTERY VOLTAGE (11–14 V)	MT-11
2	Neutral: Continuity Yes. Except neutral: Continuity No	MT-11

F14 ↔ F229

Knock path — knock sub-harness intermediate · signa



Pin	Value (conditions)	Source
2	Approximately 2.5 V	EC-127
1	Idle: body ground	

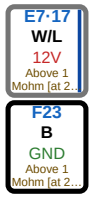
Actuators

F9
Starter motor — GY/1 · SC-13



Pin	Value (conditions)	Source
1	Battery voltage when the ignition switch is in START position	SC-18

F16
Ignition condenser — W/2 · PG-55 · EC-689



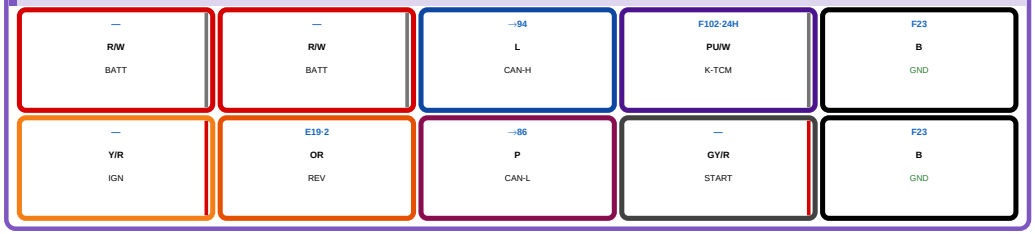
Pin	Value (conditions)	Source
1	Above 1 Mohm [at 25 °C (77 °F)]	EC-700
2	Above 1 Mohm [at 25 °C (77 °F)]	EC-700

F24
A/C compressor — B/1 · ATC-80



Pin	Value (conditions)	Source
1	When applying battery voltage direct current to the terminal, an operation sound should be heard from the magnet clutch	ATC-81

F6
A/T assembly — GY/10 · TCM · AT-186/114



Reverse swit
F36 · M/T · LT-183 · 2



Pin	Value (conditions)	Source
1	Reverse: Continuity Yes. Except reverse: Continuity No	MT-11
2	Reverse: Continuity Yes. Except reverse: Continuity No	MT-11

VTC
intake B1 — intake solenoid RH



Pin	Value (conditions)	Source
1	Between 1 and 2: 7.0–7.5 ohm [at 20 °C (68 °F)]. Between 1 or 2 and ground: ∞ ohm (Continuity should not exist) · Idle (warm-up condition): BATTERY VOLTAGE (11–14 V When revving the engine up to 2,000 rpm quickly (warm-up condition): 7–12 V	EC-182
2	Between 1 and 2: 7.0–7.5 ohm [at 20 °C (68 °F)]. Between 1 or 2 and ground: ∞ ohm (Continuity should not exist)	EC-182

VTC
intake B2 — intake solenoid LH



Pin	Value (conditions)	Source
1	Between 1 and 2: 7.0–7.5 ohm [at 20 °C (68 °F)]. Between 1 or 2 and ground: ∞ ohm (Continuity should not exist) · Idle (warm-up condition): BATTERY VOLTAGE (11–14 V When revving the engine up to 2,000 rpm quickly (warm-up condition): 7–12 V	EC-182
2	Between 1 and 2: 7.0–7.5 ohm [at 20 °C (68 °F)]. Between 1 or 2 and ground: ∞ ohm (Continuity should not exist)	EC-182

VTC
exhaust B1 — Rev-Up · exhaust RH



Pin	Value (conditions)	Source
4	9.0–11.0 ohm [at 20 °C (68 °F)]	EC-188
3	9.0–11.0 ohm [at 20 °C (68 °F)] · Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · Engine speed above 1,500 rpm (warm-up condition): 7–12 V	EC-188

Actuators

VTC

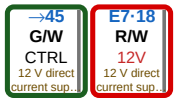
exhaust B2 — Rev-Up · exhaust LH



Pin	Value (conditions)	Source
1	9.0–11.0 ohm [at 20 °C (68 °F)]	EC-188
2	9.0–11.0 ohm [at 20 °C (68 °F)] · Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · Engine speed above 1,500 rpm (warm-up condition): 7–12 V	EC-188

EVAP

Purge solenoid — solenoid



Pin	Value (conditions)	Source
1	12 V direct current supply between terminals 1 and 2: Air passage continuity between A and B, Yes. No supply: Air passage, No	EC-363
2	12 V direct current supply between terminals 1 and 2: Air passage continuity between A and B, Yes. No supply: Air passage, No BATTERY VOLTAGE (11–14 V) at idle (accelerator pedal not depressed even slightly, after engine starting) and at about 2,000 rpm (more than 100 seconds after starting the engine)	EC-363

EVAP vent

canister vent · ECM 11



Pin	Value (conditions)	Source
1	12 V direct current supply between terminals 1 and 2: Air passage continuity between A and B, No. OFF (no supply): Air passage, Yes. Operation takes less than 1 second	EC-373
2	12 V direct current supply between terminals 1 and 2: Air passage continuity between A and B, No. OFF (no supply): Air passage, Yes. Operation takes less than 1 second · BATTERY VOLTAGE (11–14 V)	EC-373

Fuel pump

B27 · EC-710



Pin	Value (conditions)	Source
1	0.2–5.0 ohm [at 25 °C (77 °F)]	EC-715
3	0.2–5.0 ohm [at 25 °C (77 °F)]	EC-715

Coils

Coil 1

ECM62 Y/R



Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-130
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

Coil 2

ECM81 G/B



Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-132
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

Coil 3

ECM61 L/R



Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-130
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

Coil 4

ECM80 GY



Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-132
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

Coil 5

ECM60 PU/W



Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-130
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

Coil 6

ECM79 GY/R

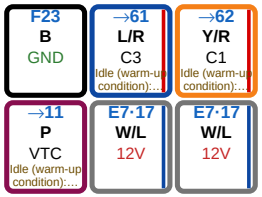


Pin	Value (conditions)	Source
1	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm. Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-132
2	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699
3	Between 1 and 2: Except 0 or ∞. Between 1 and 3, and between 2 and 3: Except 0. [at 25 °C (77 °F)]	EC-699

F18 ↔ F201

Coils B1 + VTC

6-pin · Bank 1 only · EC-691/455



Pin	Value (conditions)	Source
2	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-130
3	Idle (warm-up condition): 0–0.2 V · 2,500 rpm (warm-up condition): 0.1–0.4 V · The pulse cycle changes depending on rpm.	EC-130
4	Idle (warm-up condition): BATTERY VOLTAGE (11–14 V · When revving the engine up to 2,000 rpm quickly (warm-up condition): 7–12 V	EC-126

Injectors

Inj 1

ECM23 R/B



Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127

Inj 2

ECM42 B/R



Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128

Inj 3

ECM22 R/Y



Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127

Inj 4

ECM41 W/B



Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128

Inj 5

ECM21 SB



Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127

Inj 6

ECM40 LG

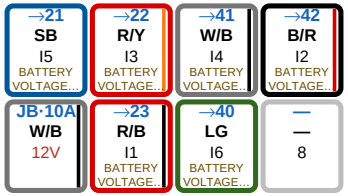


Pin	Value (conditions)	Source
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128

F33 ↔ F221

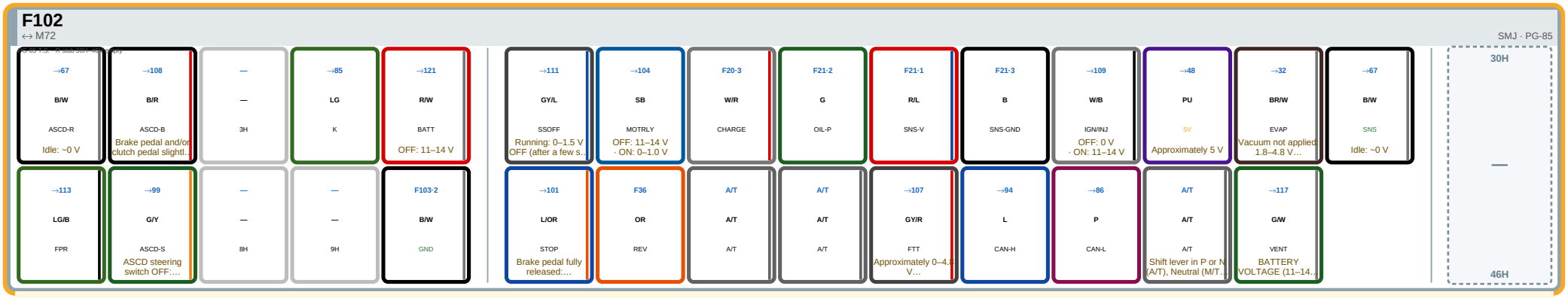
Injector junction

8-pin · front of collector · not F18



Pin	Value (conditions)	Source
1	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
2	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
3	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128
4	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128
6	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-127
7	BATTERY VOLTAGE (11–14 V) at idle (warm-up condition) and at 2,000 rpm (warm-up condition). The pulse cycle changes depending on rpm.	EC-128

Intermediates



Pin	Value (conditions)	Source
1H	Idle: ~0 V	
2H	Brake pedal and/or clutch pedal slightly depressed (M/T), or brake pedal slightly depressed (A/T): Approximately 0 V · Brake and clutch pedals fully released (M/T), or brake pedal fully released (A/T): BATTERY VOLTAGE (11–14 V)	EC-133
5H	OFF: 11–14 V	
7H	ASCD steering switch OFF: Approximately 4 V · MAIN switch pressed: Approximately 0 V · CANCEL switch pressed: Approximately 1 V · RESUME/ACCELERATE switch pressed: Approximately 3 V · SET/COAST switch pressed: Approximately 2 V	EC-133
11H	Running: 0–1.5 V · OFF (after a few s): 11–14 V	
12H	OFF: 11–14 V · ON: 0–1.0 V	
17H	OFF: 0 V · ON: 11–14 V	
18H	Approximately 5 V	EC-129
19H	Vacuum not applied: 1.8–4.8 V · At –26.7 kPa (–200 mmHg, –7.87 inHg): 2.1 to 2.5 V lower than the above value	EC-376
20H	Idle: ~0 V	
21H	Brake pedal fully released: Approximately 0 V · Brake pedal slightly depressed: BATTERY VOLTAGE (11–14 V)	EC-133
25H	Approximately 0–4.8 V · Output voltage varies with fuel tank temperature	EC-133
28H	Shift lever in P or N (A/T), Neutral (M/T): Approximately 0 V · Except above position: BATTERY VOLTAGE (11–14 V)	EC-133
29H	BATTERY VOLTAGE (11–14 V)	EC-134

E108 ↔ M15

SMJ engine room↔body

SMJ 69G · PG-84 · E108 T.S.



Pin	Value (conditions)	Source
3G	Brake pedal and/or clutch pedal slightly depressed (M/T), or brake pedal slightly depressed (A/T): Approximately 0 V · Brake and clutch pedals fully released (M/T), or brake pedal fully released (A/T): BATTERY VOLTAGE (11–14 V)	EC-133
30G	OFF: 11–14 V · ON: 0–1.0 V	
38G	Brake pedal fully released: Approximately 0 V · Brake pedal slightly depressed: BATTERY VOLTAGE (11–14 V)	EC-133
68G	OFF: 11–14 V	
69G	Running: 0–1.5 V · OFF (after a few s): 11–14 V	

B1 ↔ M12

SMJ body

69J SMJ · pin 37J · PG-84



Pin	Value (conditions)	Source
36J	Approximately 0–4.8 V · Output voltage varies with fuel tank temperature	EC-133
37J	BATTERY VOLTAGE (11–14 V)	EC-134
38J	Idle: ~0 V	
39J	Vacuum not applied: 1.8–4.8 V · At –26.7 kPa (–200 mmHg, –7.87 inHg): 2.1 to 2.5 V lower than the above value	EC-376
40J	Approximately 5 V	EC-129

Intermediates

E10 ↔ F1

Engine room ↔ control

9-pin · START · REV · A/C · SC-13/LT-1



Pin	Value (conditions)	Source
5	Approximately 5 V	EC-718
8	Idle: ~0 V	
9	1.0–4.0 V	EC-718

E12 ↔ F3

A/F + VB path

8-pin GY · VB 119/120 · IGN · GND E17



Pin	Value (conditions)	Source
4	ON: 11–14 V	
7	ON: 11–14 V	
8	ON: 11–14 V	

F103 ↔ F151

ECM ground

W 4-pin · 2/3/4 → ECM 115/1/116 → F152

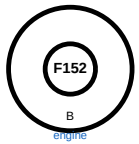


Pin	Value (conditions)	Source
2	Idle: body ground	
3	Idle: body ground	
4	Idle: body ground	

Body ground

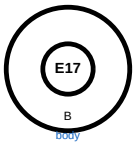
F152

ground point F152 · EC-169



Body ground

FSM: Body ground · gro



ECM F101

CAN

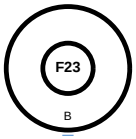
ECM harness connector · CAN-H/L · FSM



F23

Engine ground

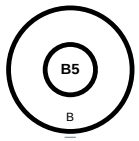
PG-35 · engine ground



B5

Body ground

PG-37 · body ground

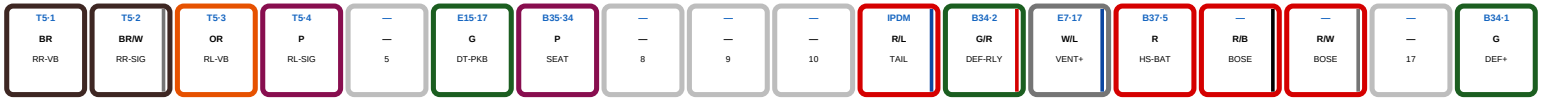


Intermediates

E106

↔ B2

18-pin W · cabin/engine · 11 tail R/L



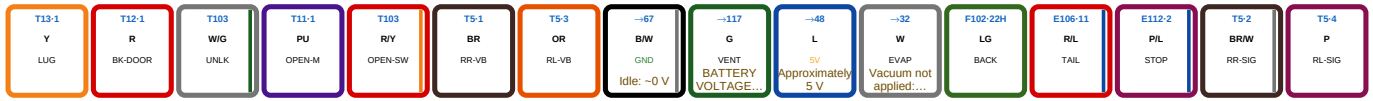
B43 ↔ T1

6-pin W · 1/6 GND · 2



T2 ↔ B44

16-pin W · 9 vent · 12



Pin	Value (conditions)	Source
8	Idle: ~0 V	
9	BATTERY VOLTAGE (11–14 V)	EC-134
10	Approximately 5 V	EC-129
11	Vacuum not applied: 1.8–4.8 V · At –26.7 kPa (–200 mmHg, –7.87 inHg): 2.1 to 2.5 V lower than the above value	EC-376

Clock spring

comb. sw M23 · 34 SIG→



Pin	Value (conditions)	Source
34	ASCD steering switch OFF: Approximately 4 V · MAIN switch pressed: Approximately 0 V · CANCEL switch pressed: Approximately 1 V · RESUME/ACCELERATE switch pressed: Approximately 3 V · SET/COAST switch pressed: Approximately 2 V	EC-133
33	Idle: ~0 V	

DLC

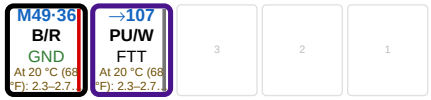
OBD-II

under dash · 16-pin J1962 · K-line ECM



Fuel tank te

fuel level sensor unit



Pin	Value (conditions)	Source
4	At 20 °C (68 °F): 2.3–2.7 kohm · At 50 °C (122 °F): 0.79–0.90 kohm · Approximately 0–4.8 V · Output voltage varies with fuel tank temperature	EC-293
5	At 20 °C (68 °F): 2.3–2.7 kohm · At 50 °C (122 °F): 0.79–0.90 kohm · Empty (*1, float rod in contact with stopper, 30 mm / 1.18 in): Approx. 80 ohm · Full (*2, float rod in contact with stopper, 210 mm / 8.27 in): Approx. 3 ohm	EC-293

Unified mete

M49

M49 · GY · DI-9



Fuel level s

B28 · B28 · 1 W/B signal · 2 Y to main · DI-



Pin	Value (conditions)	Source
1	Full (*1, float rod in contact with stopper, 8 mm / 0.31 in): Approx. 3 ohm Empty (*2, float rod in contact with stopper, 175 mm / 6.89 in): Approx. 43 ohm	DI-24
2	Full (*1, float rod in contact with stopper, 8 mm / 0.31 in): Approx. 3 ohm Empty (*2, float rod in contact with stopper, 175 mm / 6.89 in): Approx. 43 ohm	DI-24

Triple meter

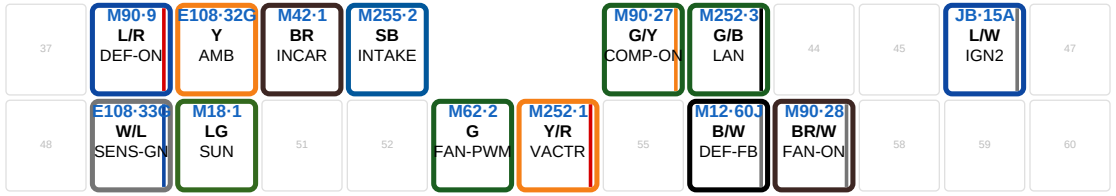
M44 · W/12 · DI-34



Pin	Value (conditions)	Source
9	Approx. 5 V	DI-45
8	Ignition switch ON (engine stopped): Approx. 1 V · Engine running [when the oil pressure is 80 psi (500 kPa)]: Approx. 3 V	DI-45

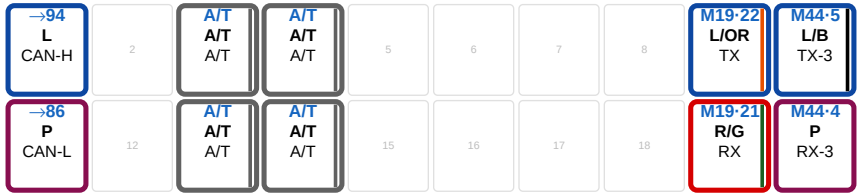
Unified mete

M50 · M50 · W/24 · ATC-44



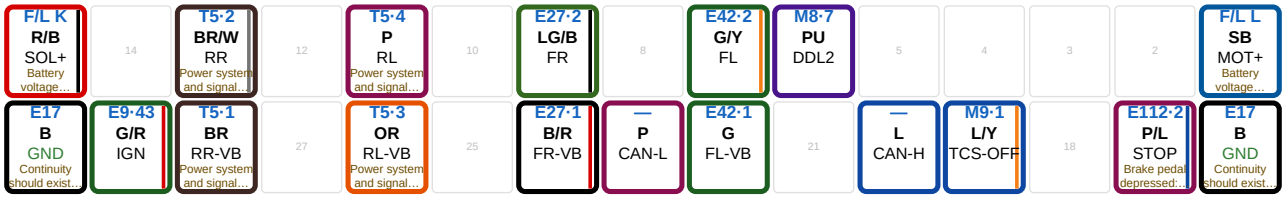
Unified mete

M48 · M48 · GY/20 · DI-9



ABS unit (ac

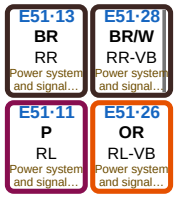
E51 · E51 · GY/30 · BRC-16/5



Pin	Value (conditions)	Source
26	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
11	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
28	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
13	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
1	Battery voltage (approx. 12V)	BRC-33
15	Battery voltage (approx. 12V)	BRC-33
16	Continuity should exist to ground	BRC-34
30	Continuity should exist to ground	BRC-34
17	Brake pedal depressed: Battery voltage (approx. 12V) · Brake pedal not depressed: 0V	BRC-35

Rear wheel s

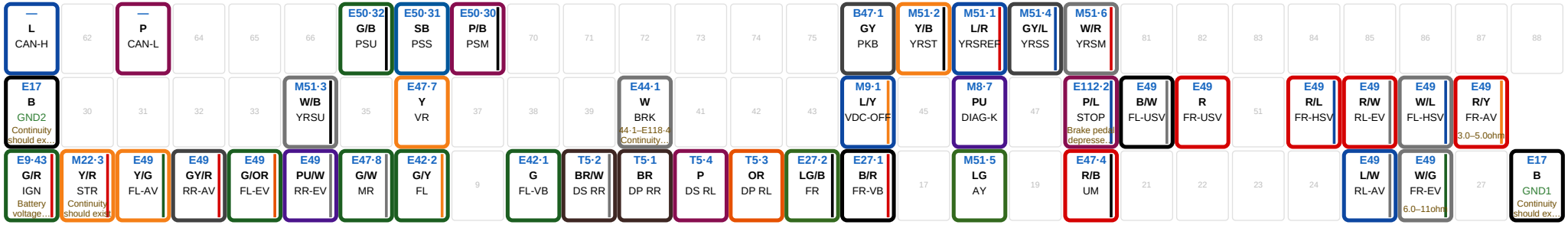
T5 · GY/4 · BRC-16



Pin	Value (conditions)	Source
1	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
2	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
3	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30
4	Power system and signal system: Continuity should exist. Ground system (actuator signal or power terminals to E51 grounds 16 and 30): Continuity should not exist	BRC-30

VDC/TCS/ABS

E118 · B/88 · BRC-100



Pin	Value (conditions)	Source
1	Battery voltage (Approx. 12V)	BRC-136
48	Brake pedal depressed: Battery voltage (approx. 12V · Brake pedal not depressed: Approx. 0V	BRC-135
40	E44·1–E118·40: Continuity should exist. E118·40 to ground: Continuity should not exist. E44·2 to ground: Continuity should exist	BRC-138
28	Continuity should exist to ground	BRC-136
29	Continuity should exist to ground	BRC-136
2	Continuity should exist	BRC-125
55	3.0–5.0ohm	BRC-129
26	6.0–11ohm	BRC-129

TCS OFF / VD

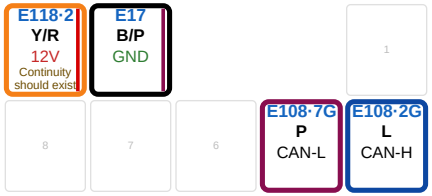
M9 · GY/6 · BRC-52/100



Pin	Value (conditions)	Source
1	Continuity should exist when pushing the switch. Continuity should not exist when releasing the switch	BRC-138
2	Continuity should exist when pushing the switch. Continuity should not exist when releasing the switch	BRC-138

Steering ang

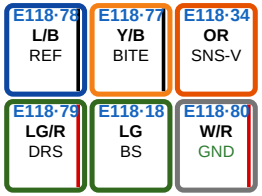
M22 · W/8 · BRC-103



Pin	Value (conditions)	Source
3	Continuity should exist	BRC-125

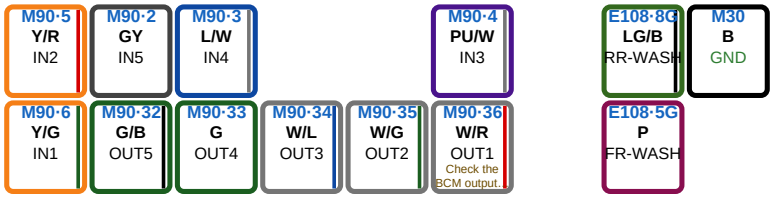
Yaw rate / s

M51 · B/6 · BRC-101



Combination

M29 · W/16 · LT-171



Pin	Value (conditions)	Source
1	Check the BCM output terminal voltage waveform of the suspect system (figure SKIA4976E)	LT-176

Remote keyle

M78 · W/4 · BL-139



Pin	Value (conditions)	Source
2	Continuity should exist. M78·2 to ground: Continuity should not exist	BL-94
4	Continuity should exist. M78·4 to ground: Continuity should not exist	BL-93

Tire pressur

M79 · W/2 · WT-14



Security ind

M34 · W/2 · BL-139



Pin	Value (conditions)	Source
1	Battery voltage	BL-160

Key switch

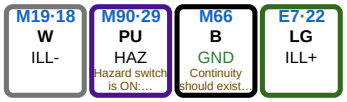
M25 · BR/2 · BL-69



Pin	Value (conditions)	Source
1	Key is inserted in ignition key cylinder: Continuity Yes. Key is removed from ignition key cylinder: Continuity No	BL-41
2	Key is inserted in ignition key cylinder: Continuity Yes. Key is removed from ignition key cylinder: Continuity No	BL-41

Hazard switc

M153 · W/4 · LT-152



Pin	Value (conditions)	Source
2	Hazard switch is ON: Approx. 0V · Hazard switch is OFF: Approx. 5V Hazard switch is ON: Continuity Yes. Hazard switch is OFF: Continuity No · Continuity should exist	LT-166
1	Continuity should exist to ground Hazard switch is ON: Continuity Yes. Hazard switch is OFF: Continuity No	LT-167

Body control

M90

M90 · W · PG-83

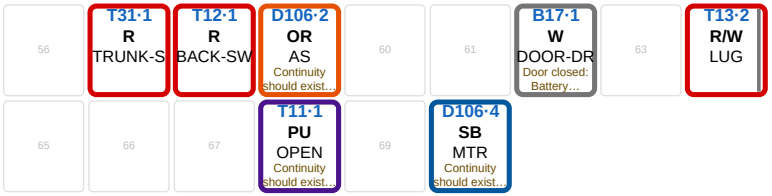


Pin	Value (conditions)	Source
12	Continuity should exist. Terminal 1 to body ground: Continuity should not exist	GW-44
19	Continuity should exist. M78·4 to ground: Continuity should not exist	BL-93
20	Continuity should exist. M78·2 to ground: Continuity should not exist	BL-94
29	Continuity should exist	LT-166
30	Back door opener switch ON: 0V · Switch OFF: Battery voltage (when interior lamp battery saver control is in OFF position: Approx. 5V)	BL-50
37	Key is inserted: Battery voltage. Key is removed: 0V	BL-41
38	Ignition switch OFF: Approx. 0V · Ignition switch ON: Battery voltage	WW-16

Body control

B83

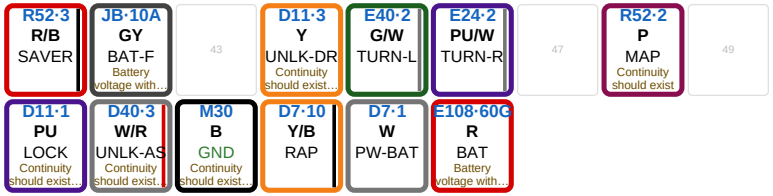
B83 · W · PG-83



Pin	Value (conditions)	Source
59	Continuity should exist. B83·59 to ground: Continuity should not exist	WW-49
62	Door closed: Battery voltage. Door open: 0V · With the door switch disconnected, 62 to ground: Battery voltage (BCM output) · Continuity should exist. Terminal 1 to body ground: Continuity should not exist	GW-44
68	Continuity should exist in both checks	BL-52
70	Continuity should exist. B83·70 to ground: Continuity should not exist · T19·1–B83·70: Continuity should exist. T19·1 to ground: Continuity should not exist. T19·2 to ground: Continuity should exist	WW-48

Body control

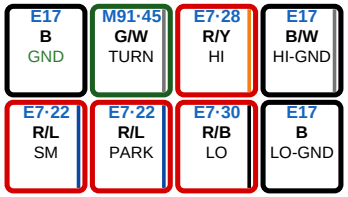
M91 · B · PG-83



Pin	Value (conditions)	Source
44	Continuity should exist. M91·44 and ·50 to ground: Continuity should not exist. At M91, when the driver door lock/unlock switch is turned to UNLOCK, 44 (Y) goes 0V → battery voltage → 0V; when turned to LOCK, 50 (PU) does the same	BL-46
48	Continuity should exist	LT-233
42	Battery voltage with the ignition switch OFF and ON	WW-16
50	Continuity should exist. M91·44 and ·50 to ground: Continuity should not exist. At M91, when the driver door lock/unlock switch is turned to UNLOCK, 44 (Y) goes 0V → battery voltage → 0V; when turned to LOCK, 50 (PU) does the same · Continuity should exist. M91·50 and ·51 to ground: Continuity should not exist. At M91, when the door lock/unlock switch is turned to LOCK, 50 (PU) goes 0V → battery voltage → 0V; when turned to UNLOCK, 51 (W/R) does the same	BL-46
51	Continuity should exist. M91·50 and ·51 to ground: Continuity should not exist. At M91, when the door lock/unlock switch is turned to LOCK, 50 (PU) goes 0V → battery voltage → 0V; when turned to UNLOCK, 51 (W/R) does the same	BL-47
52	Continuity should exist to ground	WW-16
55	Battery voltage with the ignition switch OFF and ON	WW-16

Front combin

E40 · GY/8 · LT-153



Front combin

E41 · GY/6 · LT-153



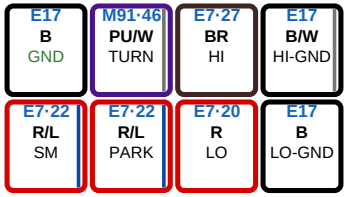
Rear combina

T10 · SB/4 · LT-153



Front combin

E24 · GY/8 · LT-154



Front combin

E25 · GY/6 · LT-154



Rear combina

T18 · SB/4 · LT-154



Rear combina

T9 · GY/3 · LT-179



Rear combina

T17 · GY/3 · LT-179



High-mounted

T202 · BR/2 · LT-180



High-mounted

D103 · BR/2 · LT-179



Map lamp (Co

R52 · W/3 · LT-220



Pin	Value (conditions)	Source
2	Map lamp switch is DOOR: Continuity Yes. Map lamp switch is OFF: Continuity No · Continuity should exist	LT-233
3	Map lamp switch is DOOR: Continuity Yes. Map lamp switch is OFF: Continuity No	LT-233

Map lamp (Ro

R53 · W/4 · LT-224



Luggage room

T13 · GY/2 · LT-220

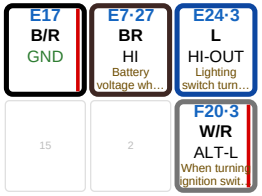


Pin	Value (conditions)	Source
1	Battery voltage. T13·1 to ground (short circuit check): Continuity should not exist	LT-236

Daytime ligh

E14

E14 · GY · LT-117



Pin	Value (conditions)	Source
5	Battery voltage when turning lighting switch to HI BEAM	LT-81
6	Lighting switch turned to the 2ND position with HI BEAM or FLASH TO PASS: Battery voltage. Daytime light operation (parking brake released, engine running, lighting switch OFF): Approx. 6V	LT-81
1	When turning ignition switch to ON: Approx. 0V · When engine is running: Battery voltage. When turning ignition switch to OFF: Approx. 0V	LT-81

Daytime ligh

E15

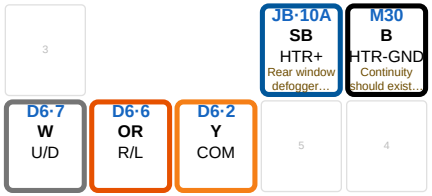
E15 · GY · LT-117



Pin	Value (conditions)	Source
9	Lighting switch turned to the 2ND position with HI BEAM or FLASH TO PASS: Approx. 0V · Daytime light operation (parking brake released, engine running, lighting switch OFF): Approx. 6V	LT-81
17	When parking brake is released: Battery voltage. When parking brake is applied: Approx. 0V	LT-81

Door mirror

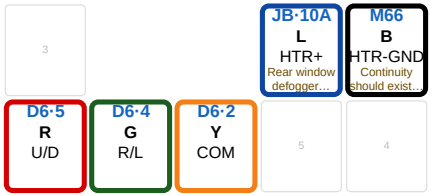
D2 · W/8 · GW-63



Pin	Value (conditions)	Source
1	Continuity should exist to ground	GW-80
2	Rear window defogger switch ON: Battery voltage. Switch OFF: 0V	GW-80

Door mirror

D32 · W/8 · GW-63



Pin	Value (conditions)	Source
1	Continuity should exist to ground	GW-81
2	Rear window defogger switch ON: Battery voltage. Switch OFF: 0V	GW-80

Passenger do

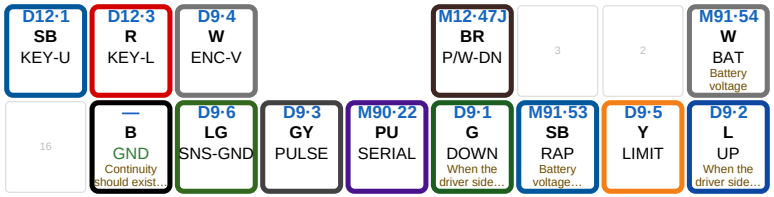
D40 · SB/4 · BL-29



Pin	Value (conditions)	Source
3	Continuity should exist. M91·50 and ·51 to ground: Continuity should not exist. At M91, when the door lock/unlock switch is turned to LOCK, 50 (PU) goes 0V → battery voltage → 0V; when turned to UNLOCK, 51 (W/R) does the same	BL-47
1	Continuity should exist. M91·50 and ·51 to ground: Continuity should not exist. At M91, when the door lock/unlock switch is turned to LOCK, 50 (PU) goes 0V → battery voltage → 0V; when turned to UNLOCK, 51 (W/R) does the same	BL-47

Power window

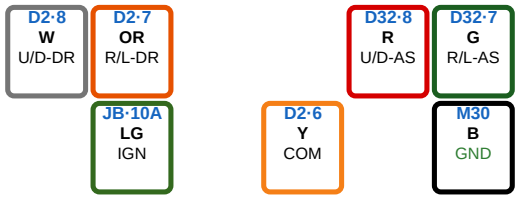
D7 · W/16 · GW-24



Pin	Value (conditions)	Source
1	Battery voltage	GW-47
8	When the driver side power window motor is in UP operation: Battery voltage at terminal 8. In DOWN operation: Battery voltage at terminal 11	GW-27
10	Battery voltage (ignition ON, and also within 45 seconds after the ignition switch is turned to OFF)	GW-47
11	When the driver side power window motor is in UP operation: Battery voltage at terminal 8. In DOWN operation: Battery voltage at terminal 11	GW-27
15	Continuity should exist to ground. Reference value: Approx. 0V	GW-47

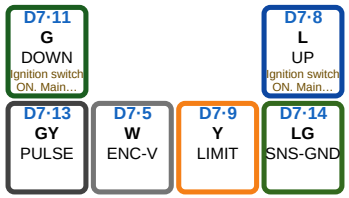
Door mirror

D6 · W/10 · GW-84



Power window

D9 · W/6 · GW-24



Pin	Value (conditions)	Source
1	Ignition switch ON. Main switch UP: D9·1 approx. 0V, D9·2 battery voltage. Main switch DOWN: D9·1 battery voltage, D9·2 approx. 0V	GW-34
2	Ignition switch ON. Main switch UP: D9·1 approx. 0V, D9·2 battery voltage. Main switch DOWN: D9·1 battery voltage, D9·2 approx. 0V	GW-34

Door key cyl

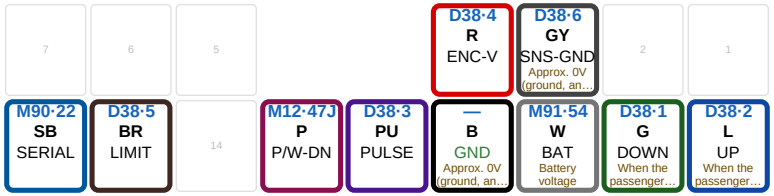
D12 · BR/3 · GW-24



Pin	Value (conditions)	Source
1	1–2: Neutral/Lock continuity No, Unlock continuity Yes. 3–2: Neutral/Unlock continuity No, Lock continuity Yes. Also D12·2 to ground: Continuity should exist	GW-50
2	1–2: Neutral/Lock continuity No, Unlock continuity Yes. 3–2: Neutral/Unlock continuity No, Lock continuity Yes. Also D12·2 to ground: Continuity should exist	GW-50
3	1–2: Neutral/Lock continuity No, Unlock continuity Yes. 3–2: Neutral/Unlock continuity No, Lock continuity Yes. Also D12·2 to ground: Continuity should exist	GW-50

Power window

D37 · W/16 · GW-25



Pin	Value (conditions)	Source
3	Approx. 0V (ground, and limit switch/encoder ground)	GW-28
8	When the passenger side power window motor is in UP operation: Battery voltage at terminal 8. In DOWN operation: Battery voltage at terminal 9	GW-28
9	When the passenger side power window motor is in UP operation: Battery voltage at terminal 8. In DOWN operation: Battery voltage at terminal 9	GW-28
10	Battery voltage	GW-28
11	Approx. 0V (ground, and limit switch/encoder ground)	GW-28

Power window

D38 · W/6 · GW-25



Pin	Value (conditions)	Source
1	Ignition switch ON. Sub-switch closing: D38·1 approx. 0V, D38·2 battery voltage. Sub-switch opening: D38·1 battery voltage	GW-34
2	Ignition switch ON. Sub-switch closing: D38·1 approx. 0V, D38·2 battery voltage. Sub-switch opening: D38·1 battery voltage	GW-34

D1 ↔ M11

driver door SMJ

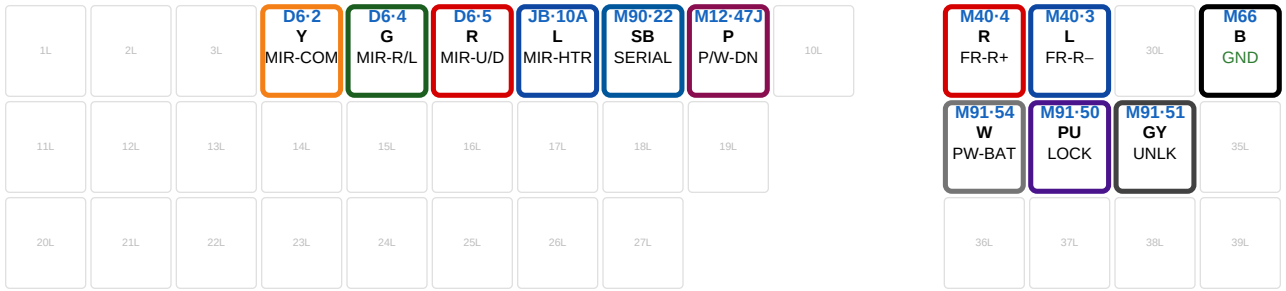
D1 · W/39 · PG-85



D31 ↔ M74

passenger door SMJ

D31 · W/39 · PG-85



Driver door

B17 · W/3 · BL-27 / BL



Pin	Value (conditions)	Source
1	Switch pushed (door closed): Continuity No. Switch released (door open): Continuity Yes Continuity should exist. Terminal 1 to body ground: Continuity should not exist	GW-44

Passenger do

B23 · W/3 · BL-27 / BL



Pin	Value (conditions)	Source
1	Switch pushed (door closed): Continuity No. Switch released (door open): Continuity Yes Continuity should exist. Terminal 1 to body ground: Continuity should not exist	GW-44

Driver door

D11 · SB/4 · BL-29 / B



Pin	Value (conditions)	Source
3	Continuity should exist. M91·44 and ·50 to ground: Continuity should not exist. At M91, when the driver door lock/unlock switch is turned to UNLOCK, 44 (Y) goes 0V → battery voltage → 0V; when turned to LOCK, 50 (PU) does the same	BL-46
1	Continuity should exist. M91·44 and ·50 to ground: Continuity should not exist. At M91, when the driver door lock/unlock switch is turned to UNLOCK, 44 (Y) goes 0V → battery voltage → 0V; when turned to LOCK, 50 (PU) does the same	BL-46

Back door sw

T12 · W/3 · BL-27



Pin	Value (conditions)	Source
1	Open position: Continuity Yes. Also T12·3 (B) to ground: Continuity should exist	BL-89
3	Open position: Continuity Yes. Also T12·3 (B) to ground: Continuity should exist	BL-89

Back door op

T103 · GY/2 · BL-30 /



Pin	Value (conditions)	Source
2	Back door opener switch ON: Continuity Yes. Switch OFF: Continuity No. T103·1 (P) – M90·30 (P): Continuity should exist	BL-50
1	Back door opener switch ON: Continuity Yes. Switch OFF: Continuity No. T103·1 (P) – M90·30 (P): Continuity should exist	BL-50

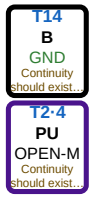
Trunk opener

B71 · W/2 · BL-125



Back door op

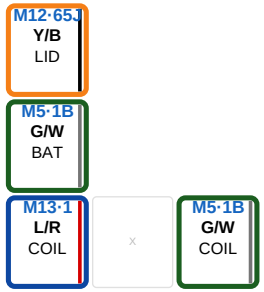
T11 · W · BL-30



Pin	Value (conditions)	Source
2	Continuity should exist in both checks	BL-52
1	Continuity should exist in both checks	BL-52

Fuel lid ope

M59 · L · BL-54



Fuel lid ope

M13 · GY · BL-54 / BL-



Pin	Value (conditions)	Source
1	ON (Pushed): Continuity Yes. OFF (Released): Continuity No. M13·2 (B) to ground: Continuity should exist	BL-60
2	ON (Pushed): Continuity Yes. OFF (Released): Continuity No. M13·2 (B) to ground: Continuity should exist	BL-60

Fuel lid ope

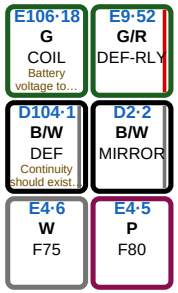
T19 · W · BL-54 / BL-5



Pin	Value (conditions)	Source
2	T19·1–B83·70: Continuity should exist. T19·1 to ground: Continuity should not exist. T19·2 to ground: Continuity should exist	BL-60
1	T19·1–B83·70: Continuity should exist. T19·1 to ground: Continuity should not exist. T19·2 to ground: Continuity should exist	BL-60

Rear defogge

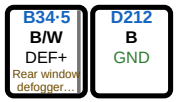
B34 · BR · GW-63 / GW-



Pin	Value (conditions)	Source
1	Battery voltage to ground (the wire colour is G or P in the FSM)	GW-76
5	Continuity should exist between B34·5 (B/W) and B202·1 (B)	GW-79

Rear glass d

D104 / D211 · B · GW-6



Pin	Value (conditions)	Source
1	Rear window defogger switch ON: Battery voltage. Switch OFF: 0V	GW-77

Rear glass d

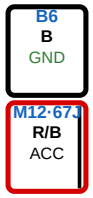
B202 / B203 · B · GW-6



Pin	Value (conditions)	Source
1	Continuity should exist between B34·5 (B/W) and B202·1 (B)	GW-79

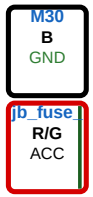
Luggage 12 V

B36 · B · WW-54



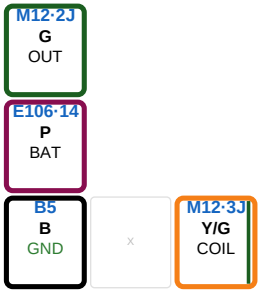
Dash 12 V so

M38 · B · WW-54



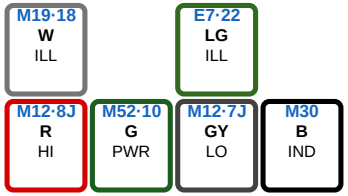
Heated seat

B37 · L · SE-59



Driver heate

M154 · W · SE-60



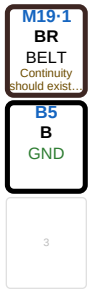
Passenger he

M155 · BR · SE-61



Driver seat

B8 · W · SRS-12



Pin	Value (conditions)	Source
1	Continuity should exist. M19·1 to ground: Continuity should not exist	DI-98

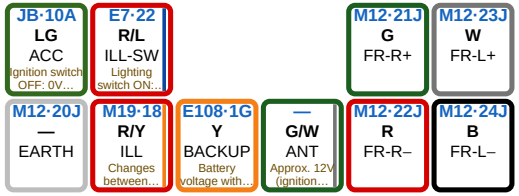
Passenger se

B11 · W · SRS-12



Audio unit M

M40 · W · AV-11



Pin	Value (conditions)	Source
10	Ignition switch OFF: 0V · ACC or ON: Battery voltage	AV-36
8	Lighting switch ON: Approx. 12V · Lighting switch OFF: Approx. 0V	AV-28
7	Changes between approx. 0 and approx. 12V when the illumination control switch is operated with the lighting switch in the ON position	AV-28
6	Battery voltage with the ignition switch OFF, ACC and ON	AV-36
5	Approx. 12V (ignition switch ON, antenna signal, output)	AV-28

Audio unit M

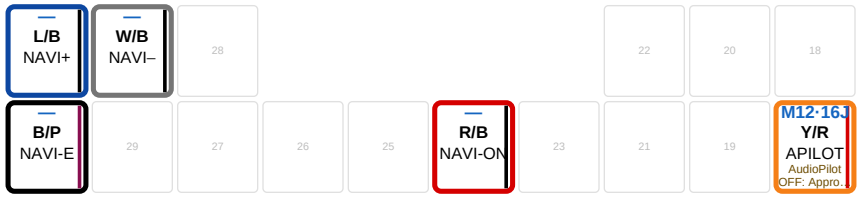
M41 · W · AV-12



Pin	Value (conditions)	Source
12	Approx. 12V (ignition switch ON, Amp. ON signal, output)	AV-29
11	Approx. 0V (ignition switch ON, shield)	AV-29

Audio unit M

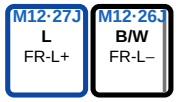
M39 · W · AV-12



Pin	Value (conditions)	Source
17	AudioPilot OFF: Approx. 7.5V · AudioPilot ON: Approx. 0V	AV-30

Driver door

D4 · W · AV-11



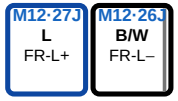
Driver door

D5 · BR · AV-17



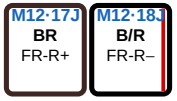
Driver tweet

D3 · BR · AV-11



Passenger do

D34 · W · AV-11



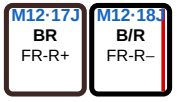
Passenger do

D35 · BR · AV-17



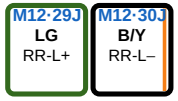
Passenger tw

D33 · BR · AV-11



Rear speaker

B40 · BR · AV-12



Rear speaker

B42 · BR · AV-12



Combination

M19 · W/24 · DI-67



Pin	Value (conditions)	Source
1	When seat belt is fastened: Approx. 12 V · When seat belt is unfastened: Approx. 0 V · Continuity should exist. M19·1 to ground: Continuity should not exist	DI-97
10	Continuity should exist to ground. Reference value (DI-11, terminal 11, ignition ON): Approx. 0 V	DI-17
14	Ignition switch OFF: 0V · ACC or ON: Battery voltage	DI-16
23	Ignition switch OFF or ACC: 0V · Ignition switch ON: Battery voltage	DI-16
24	Battery voltage with the ignition switch OFF, ACC and ON	DI-16

Brake fluid

E44 · GY/2 · DI-67



Pin	Value (conditions)	Source
1	E44·1–E118·40: Continuity should exist. E118·40 to ground: Continuity should not exist. E44·2 to ground: Continuity should exist	BRC-138
2	E44·1–E118·40: Continuity should exist. E118·40 to ground: Continuity should not exist. E44·2 to ground: Continuity should exist	BRC-138

Parking brak

B47 · B/1 · DI-67



Outside temp

E34 · B · ATC-44



Pin	Value (conditions)	Source
1	Approx. 5V to ground After disconnecting ambient sensor connector E34, measure resistance between terminals 1 and 2 at sensor side, using the table: -15 °C (5 °F): 12.73 kΩ; -10 (14): 9.92; -5 (23): 7.80; 0 (32): 6.19; 5 (41): 4.95; 10 (50): 3.99; 15 (59): 3.24; 20 (68): 2.65; 25 (77): 2.19; 30 (86): 1.81; 35 (95): 1.51; 40 (104): 1.27; 45 (113): 1.07	ATC-97
2	After disconnecting ambient sensor connector E34, measure resistance between terminals 1 and 2 at sensor side, using the table: -15 °C (5 °F): 12.73 kΩ; -10 (14): 9.92; -5 (23): 7.80; 0 (32): 6.19; 5 (41): 4.95; 10 (50): 3.99; 15 (59): 3.24; 20 (68): 2.65; 25 (77): 2.19; 30 (86): 1.81; 35 (95): 1.51; 40 (104): 1.27; 45 (113): 1.07	ATC-99

Blower motor

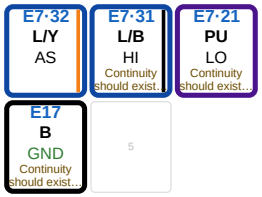
M62 · W · ATC-42



Pin	Value (conditions)	Source
2	Usual terminal 2 (G) drive signal duty ratios while varying fan speed between Lo and Hi are shown in the table on ATC-78 (figure). If the fan air flow does not change, the same table applies	ATC-78

Front wiper

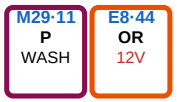
E52 · GY/5 · WW-11



Pin	Value (conditions)	Source
2	Continuity should exist. E7·31 to ground: Continuity should not exist. During HI operation: Battery voltage at E7·31 to ground	WW-27
3	Continuity should exist. E7·21 to ground: Continuity should not exist. During LO operation (ACTIVE TEST or auto active test): Battery voltage at E7·21 to ground	WW-26
4	Continuity should exist to ground	WW-23

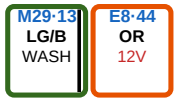
Front washer

E29 · GY/2 · WW-12



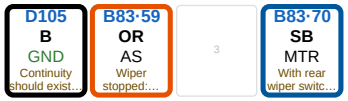
Rear washer

E28 · GY/2 · WW-40



Rear wiper m

D106 · W/4 · WW-40



Pin	Value (conditions)	Source
1	Continuity should exist to ground	WW-47
2	Wiper stopped: Battery voltage. Wiper operating: Approx. 0V · Continuity should exist. B83·59 to ground: Continuity should not exist	WW-49
4	With rear wiper switch ON: Battery voltage to ground · Continuity should exist. B83·70 to ground: Continuity should not exist	WW-47

Soft top cir

B55 · W/2 · RF-25



Soft top con

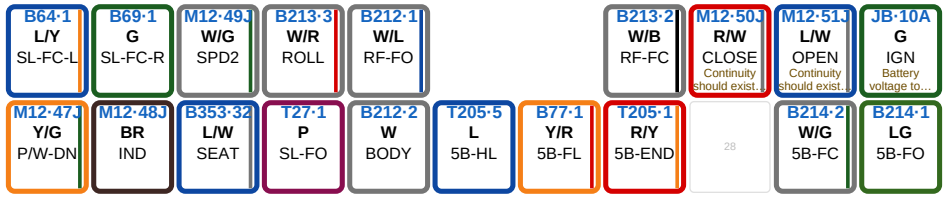
B66 · W · RF-25



Pin	Value (conditions)	Source
1	Battery voltage to ground	RF-39
7	Continuity should exist to ground	RF-39
8	Battery voltage to ground	RF-39
16	Continuity should exist to ground	RF-39

Soft top con

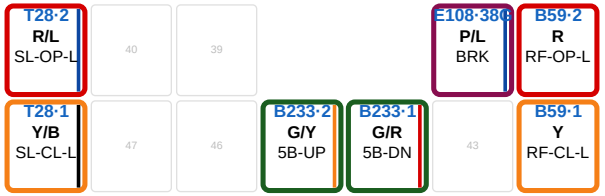
B67 · W · RF-25



Pin	Value (conditions)	Source
17	Battery voltage to ground (ignition supply)	RF-39
18	Continuity should exist. B67·18 to ground: Continuity should not exist	RF-40
19	Continuity should exist. B67·19 to ground: Continuity should not exist. With the connectors attached, B67·19 to ground: Approx. 5V (control unit output signal)	RF-42

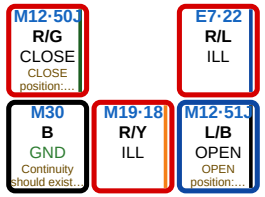
Soft top con

B68 · W · RF-25



Soft top swi

M14 · W · RF-26



Pin	Value (conditions)	Source
1	Continuity should exist to ground · OPEN position: Continuity Yes · CLOSE position: Continuity Yes	RF-40
3	OPEN position: Continuity Yes · Continuity should exist. B67·18 to ground: Continuity should not exist	RF-40
4	CLOSE position: Continuity Yes Continuity should exist. B67·19 to ground: Continuity should not exist. With the connectors attached, B67·19 to ground: Approx. 5V (control unit output signal)	RF-42

A/T selector

M47 · W · AT-240 / AT-

